FLUCONAZOLE

CYP2C9/3A4 inhibitor
QT prolongation

CHEMICAL NAME 2-(2,4-difluorophenyl)-1,3-bis(1H-1,2,4-triazol-1-yl)propan-2-ol
CLASS Triazole antifungal
TRADE NAMES Diflucan · Flucoral · Fungustatin · Triflucan

THERAPEUTIC USES / INDICATIONS

- Vaginal candidiasis (single 150 mg dose).
- Oropharyngeal and oesophageal candidiasis.
- Candidaemia and invasive candidiasis (susceptible species).
- Cryptococcal meningitis (consolidation/maintenance).
- Antifungal prophylaxis in neutropenia, transplant.

ADMINISTRATION

- **PO** with or without food — bioavailability >90% (oral = IV).
- **IV** infusion max 10 mL/min.
- Shake suspension well.

DOSAGES

- **Vaginal thrush:** 150 mg single dose.
- **Oropharyngeal:** 50–100 mg daily for 7–14 days.
- **Oesophageal:** 200–400 mg daily for 14–21 days.
- **Invasive candidiasis:** 800 mg loading, then 400 mg daily.
- **Children:** 3–12 mg/kg daily. CrCl ≤50: halve dose after loading.

PHARMACOKINETICS

- **Peak:** 1–2 h. Bioavailability >90%.
- Excellent CSF and tissue penetration; little metabolism.
- Excreted unchanged in urine. **t½:** ~30 h (once daily).

PHARMACODYNAMICS

- Inhibits fungal **lanosterol 14α-demethylase** (CYP51) → ↓ ergosterol in cell membrane → fungistatic.
- Active against most *Candida* (not *C. krusei*; reduced for *C. glabrata*) and *Cryptococcus*; no mould activity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, abdominal pain, diarrhoea.
- Headache.
- Rash.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, torsades.
- Hepatotoxicity.
- Severe skin reactions (exfoliative, SJS).
- Adrenal insufficiency (rare).

CONTRAINDICATIONS & CAUTIONS

- Azole hypersensitivity.
- Use with QT-prolonging drugs metabolised by CYP3A4 (cisapride, pimozide, erythromycin).
- **Caution:** long QT, liver disease, pregnancy (high-dose prolonged — teratogenic).

KEY INTERACTIONS

- **Warfarin:** ↑↑ INR.
- **Sulfonylureas:** hypoglycaemia; **phenytoin:** toxicity.
- **Statins (simva/atorva), ciclosporin, tacrolimus, celecoxib:** ↑ levels.
- **Rifampicin:** ↓ fluconazole.

NURSING MANAGEMENT

- Review interacting drugs, esp. warfarin.
- Monitor LFTs in prolonged therapy; ECG in at-risk patients.
- Check renal function for dosing.
- Assess oral lesions / symptoms.
- Monitor glucose with sulfonylureas.

PATIENT EDUCATION

- For thrush, one capsule is usually enough; symptoms settle in 2–3 days.
- Tell your doctor about all medicines (especially blood thinners).
- Report yellow skin, palpitations or rash.
- Avoid in pregnancy unless prescribed.

KEY POINTS

- Triazole; inhibits ergosterol synthesis.
- Excellent oral absorption and CSF penetration.
- 150 mg single dose for vaginal thrush.
- Raises INR; QT risk.